

M – Tech Production & Marketing

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REPORT OF EXPERIMENT 12

Generative AI — GANs, VAEs & Diffusion Models

Introduction

This report summarizes the activities completed during **Experiment 12** of the M-Tech Production & Marketing AI/ML Internship Program. The experiment focused on understanding the fundamentals of **Generative Artificial Intelligence (Generative AI)** by exploring three widely used generative models: **Generative Adversarial Networks (GANs)**, **Variational Autoencoders (VAEs)**, and **Diffusion Models**. Practical implementation was performed using Python to understand how these models generate synthetic data and their applications in modern AI systems.

Objectives

- To understand the concept of Generative Artificial Intelligence.
- To explore the working principles of GANs, VAEs, and Diffusion Models.
- To generate sample synthetic data using Python.
- To compare different Generative AI models.
- To understand real-world applications of Generative AI.

Task Description

The assigned task was to study the fundamentals of Generative AI and implement a simple Python program demonstrating its basic concepts. The program generated sample synthetic data using random values, introduced the three major Generative AI

models, compared their characteristics, and highlighted their practical applications in Artificial Intelligence.

Work Done

Today, I developed a Python program to understand the basic concepts of Generative AI. I generated a random latent vector to simulate the input provided to a generative model and produced sample synthetic data using NumPy. I explored the three major Generative AI models—GANs, VAEs, and Diffusion Models—and compared their working principles and capabilities. In addition, I reviewed several real-world applications of Generative AI, including image generation, image enhancement, content creation, face generation, and medical imaging. After completing the implementation, I successfully tested the program and verified all outputs.

Challenges Faced

During the experiment, understanding the differences between GANs, VAEs, and Diffusion Models required careful study because each model follows a different approach to generating synthetic data. I overcame this challenge by reviewing their characteristics, comparing their workflows, and practicing the implementation multiple times to gain a better understanding.

Key Learnings

This experiment improved my understanding of Generative Artificial Intelligence and its role in creating synthetic data. I learned the purpose of latent vectors, explored the working principles of GANs, VAEs, and Diffusion Models, and understood their practical applications in image generation, content creation, medical imaging, and other AI-based solutions. The practical implementation also strengthened my Python programming skills and enhanced my knowledge of modern AI technologies.

Conclusion

The experiment was completed successfully, and all learning objectives were achieved. Studying Generative AI models provided valuable insight into modern Artificial Intelligence techniques used for generating realistic synthetic data. The practical implementation enhanced my understanding of Generative AI concepts and prepared me for more advanced AI and Deep Learning applications.

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